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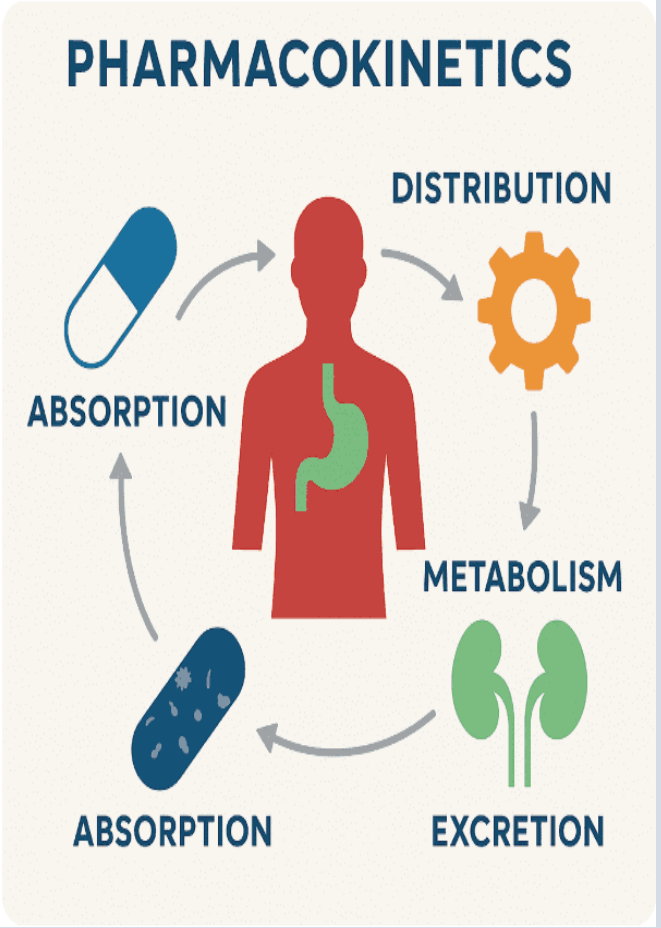
College of Medicine | Department of Pharmacology

Pharmacokinetics

Absorption & Distribution

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MSc Pharmacology



LECTURE OVERVIEW

Objectives

By the end of this lecture you should be able to:

1. Understand what is meant by Pharmacokinetics.
2. List the principal routes of drug administration.
3. Identify the effect of pH on drug absorption.
4. Describe Ion trapping & 1st-pass metabolism.
5. Define Bioavailability.

Objectives (continued)

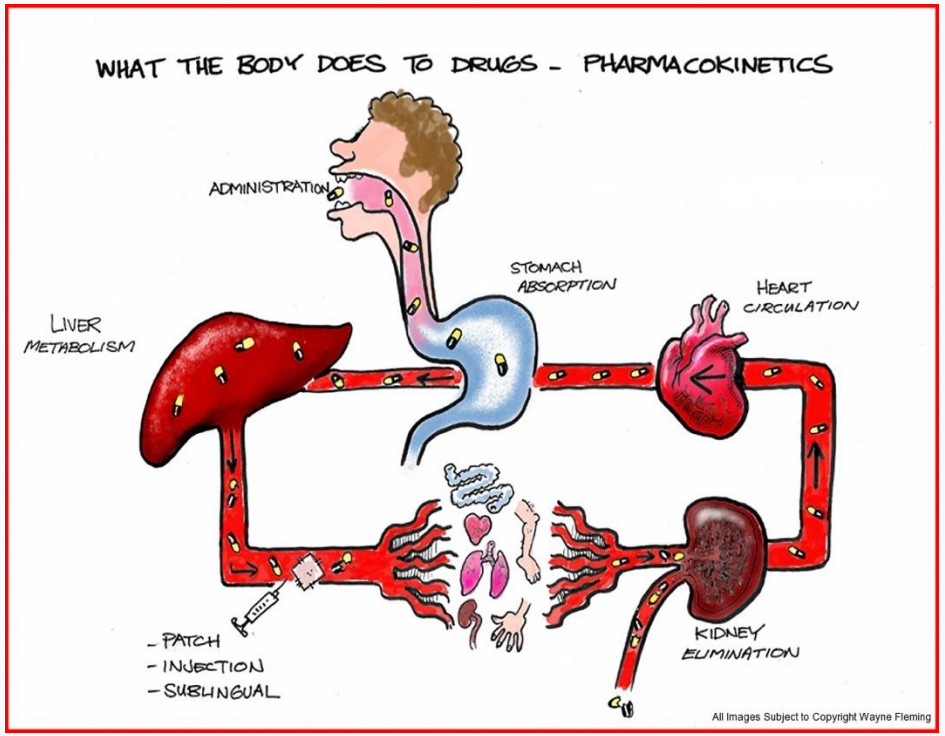
6 List factors that affect drug distribution.

7 Know the types of protein binding.

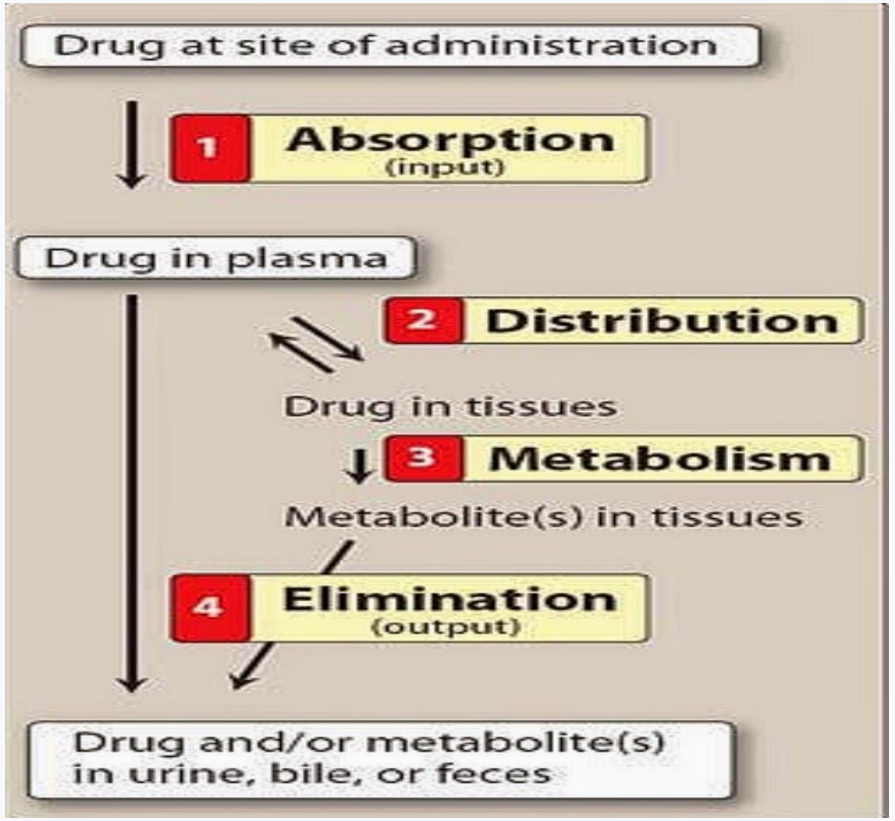
8 Define Volume of distribution.

What Is Pharmacokinetics?

Pharmacokinetics is the movement of a drug over time through the body.



The ADME journey of a drug through the body — from administration to elimination.



Pharmacokinetics

Pharmacokinetics: refers to what the body does to a drug. It includes 4 processes:

1

Absorption

2

Distribution

3

Metabolism

4

Excretion

Absorption

Definition:

It is the passage of a drug from the site of administration to the plasma.

The rate and efficacy of absorption depend on the route of administration.

With the I.V. route, once the whole dose reaches systemic circulation, absorption is complete.

Routes of Drug Administration

For each route of drug administration there is a specific rate and efficiency of drug absorption.

1. **Oral**
2. **Sublingual**
3. **Rectal**
4. **Other epithelial surfaces (conjunctiva, nose…)**
5. **Inhalation**
6. **Injection (IV, IM, SC…)**

Mechanisms of Absorption from the GI Tract

1 · Passive Diffusion

High concentration → Low concentration

- The driving force is the concentration gradient.
- Lipid-soluble drugs pass through biological membranes.
- Water soluble through aqueous pores.
- **The vast majority of drugs are absorbed by this mechanism.**

2 · Facilitated Diffusion

- Agents can enter the cell through specialized transmembrane carrier proteins that facilitate the passage of large molecules.
- No energy is required.
- Saturable.
- Can be inhibited.

Mechanisms of Absorption (continued)

3 · Active Transport

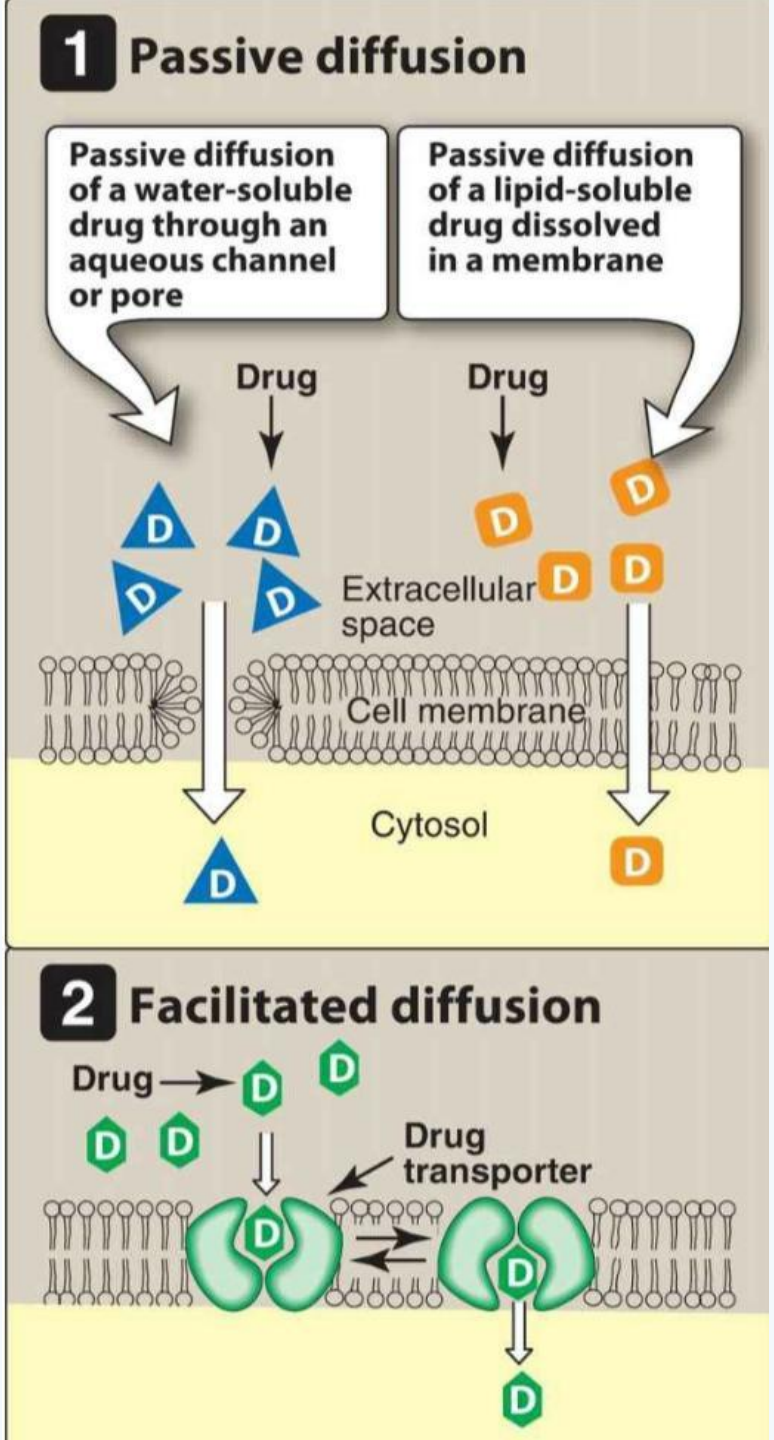
Low concentration → High concentration

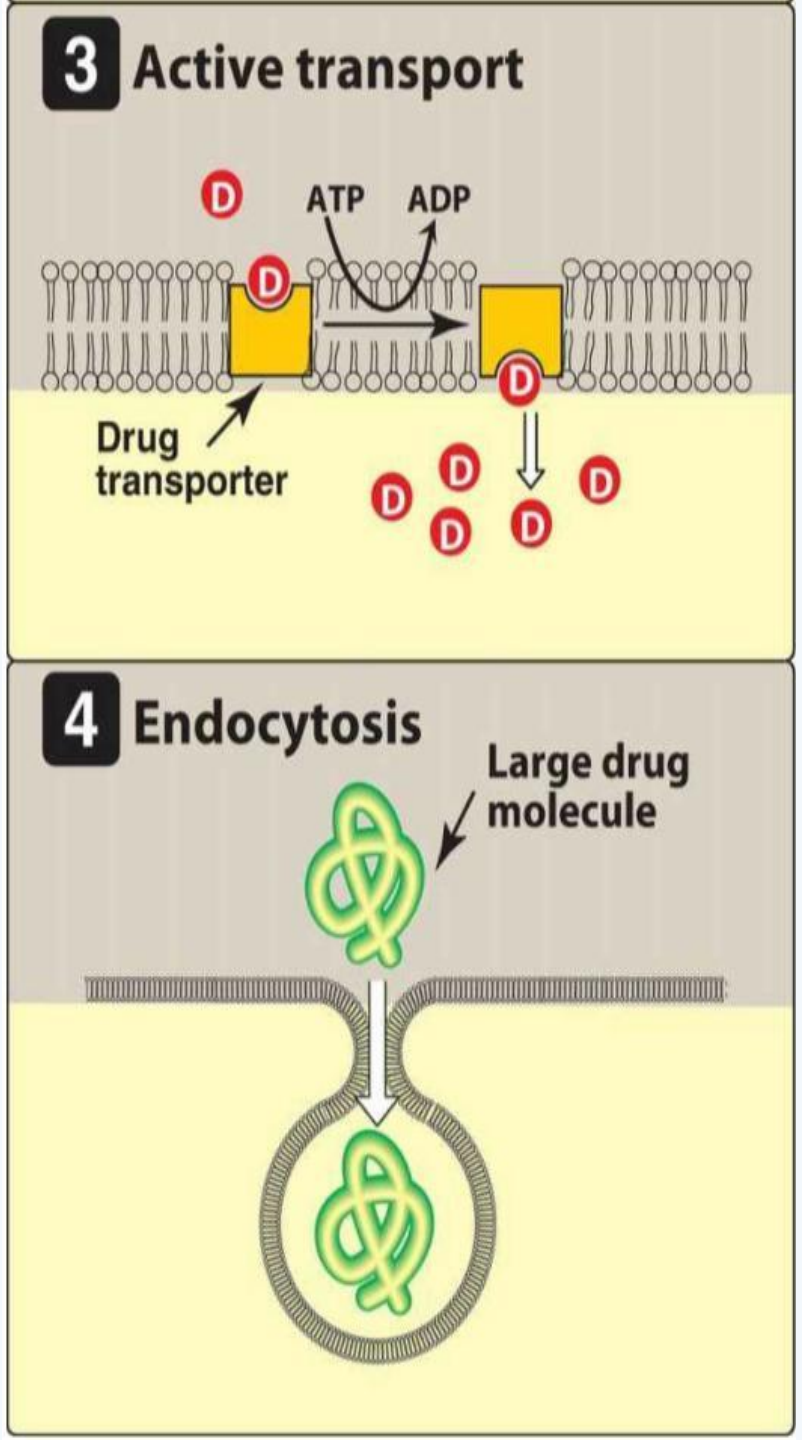
- Requires specific carrier proteins.
- Against the concentration gradient.
- This process needs energy (ATP).
- Saturable.

4 · Endocytosis & Exocytosis

- For drugs of exceptionally large size.
- Vitamin B12 — through endocytosis.
- Norepinephrine — through exocytosis.

Absorption Mechanisms





Factors Affecting Drug Absorption

A. Factors related to the drug

- **Molecular size**: small molecules are absorbed more readily than large molecules.
- **Dose:** absorption increases with increasing the dose (up to a limit).
- **Drug formulations**: e.g., sustained-release tablets are slow in absorption.

Factors Affecting Drug Absorption (continued)

- **Local effects of the drug**: drugs producing vasoconstriction (VC) ↓ their own absorption, e.g., adrenaline combined with lidocaine.
- **Drug combination**: e.g., vitamin C ↑ absorption of iron.
- **Lipid solubility**: drug ionization and the pKa of the drug.

Factors: The Absorbing Surface

B. Factors related to the absorbing surface

- **Route of administration:** i.v. Route is the fastest, while rectal is the slowest.
- **Integrity of the absorbing surface**: may ↑ or ↓ absorption.
- **Local blood flow:** ischemia ↓ absorption.
- **Specific factors**: e.g. apoferritin system for iron, etc.
- **G.I.T. motility:**
  - Rapid passage (e.g. diarrhea) → low absorption.
  - Delayed gastric emptying → delayed absorption.
  - Presence of food → diluted drug & delayed gastric emptying.

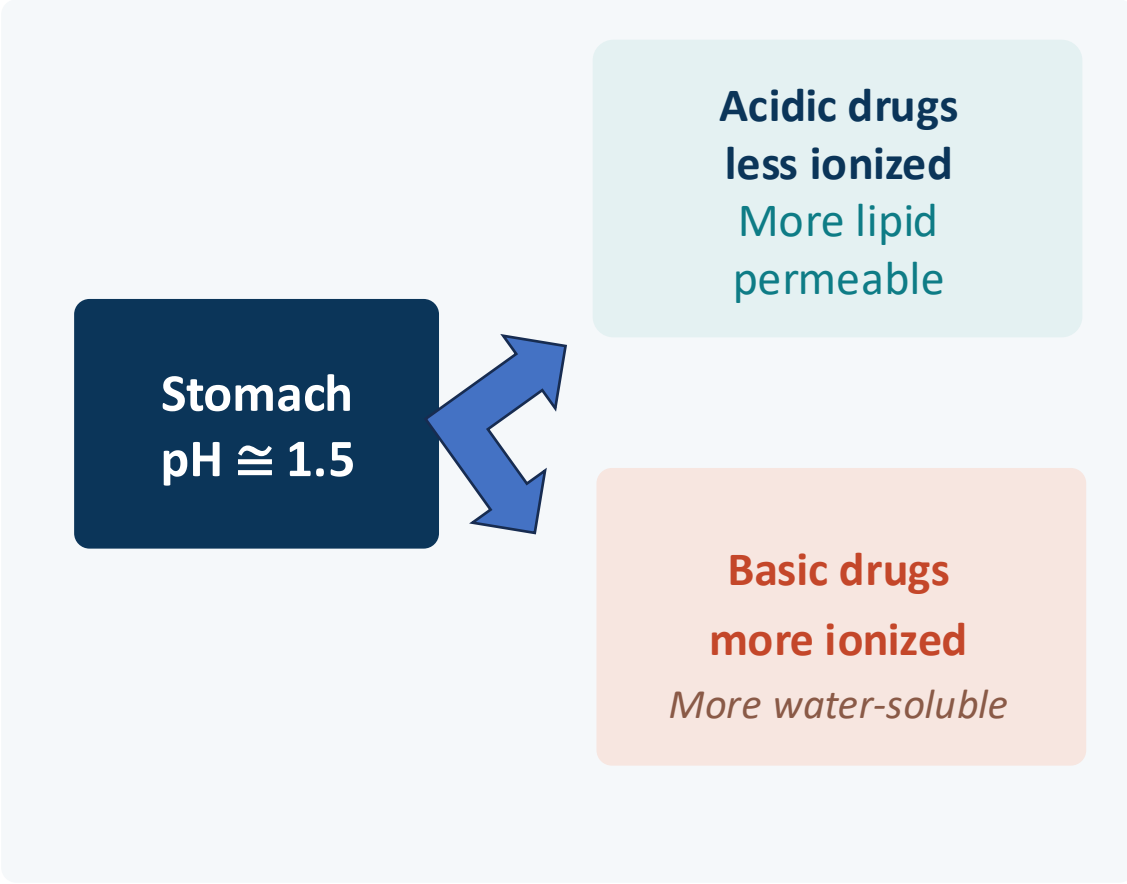
Factors: Physicochemical

C. Physicochemical factors

Drug interaction (e.g., tetracycline binding to Ca^{2+} in milk) prevents its absorption.

Effect of pH on Drug Absorption

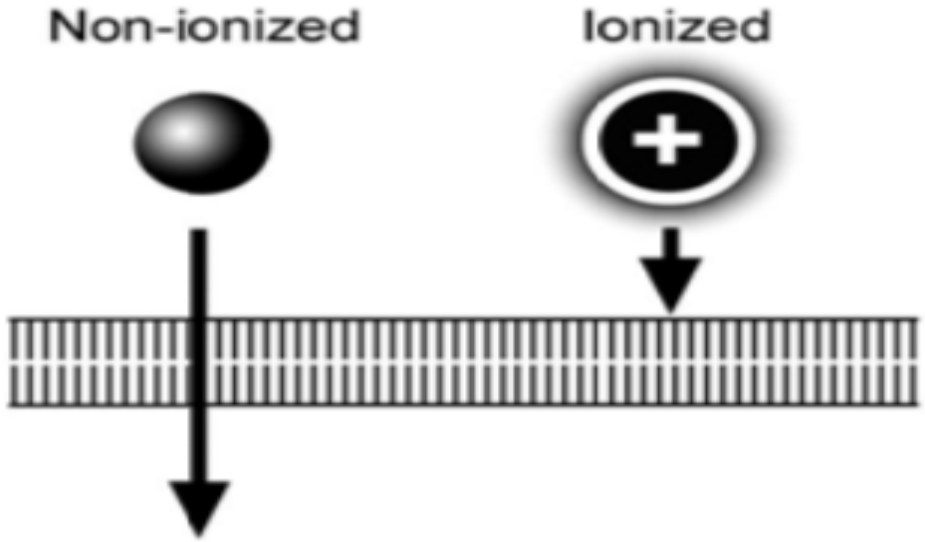
- Cell membrane permeable to non-ionized (lipid-soluble) drugs.
- Passage depends on:
  - –Degree of drug ionization.
  - –pH of the environment.



The pKa and Drug Ionization

Principles

- Ionized (polar; charged) drugs are poorly absorbed.
- Non-ionized (non-polar, non-charged) drugs are more absorbed.
- Most drugs are weak acids or bases.
- They become ionized or non-ionized according to the pH around them.
  - Acidic drugs (e.g., aspirin) are more ionized in alkaline pH.
  - Basic drugs (e.g., amphetamine) are more ionized in acidic pH.



pKa of a drug: the pH at which 50% of the drug is ionized, and 50% is non-ionized.
P=inverse log; Ka=(association/dissociation constant)

pH & Drug Kinetics: Aspirin Example

Aspirin is an acidic drug — its pKa = 3.5 | Stomach pH = 1.5 | Intestine pH = 8.5

In the Stomach

- Aspirin is acidic and becomes more absorbable at acidic pH.
- *Log (Unionized/Ionized) = pKa − pH = 3.5 − 1.5 = 2 (log 2 = 10^2)*
- **Ratio of unionized : ionized ≈ 100 : 1 — about 0.99 parts absorbed, 0.01 parts non-absorbed.**

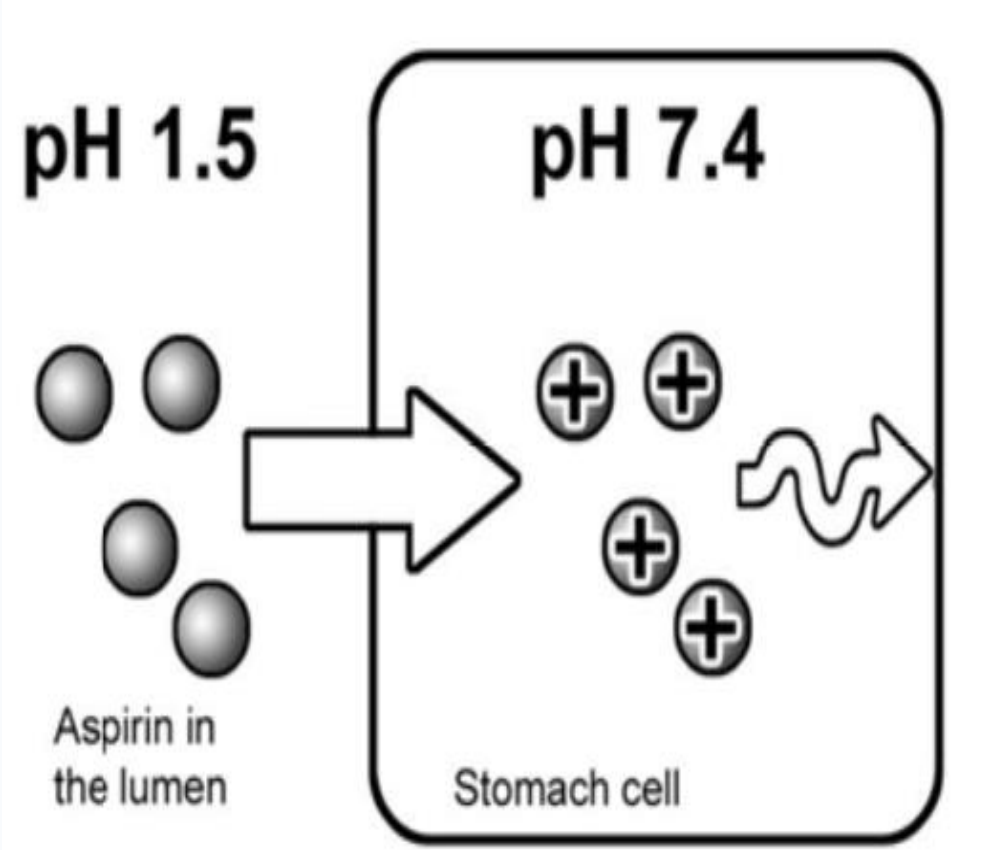
In the Intestine

- Aspirin becomes less absorbable in alkaline pH.
- *Log (Unionized/Ionized) = pKa − pH = 3.5 − 8.5 = −5 ($\log^{-5} = 10^{-5}$)*
- **Ratio of unionized : ionized ≈ 1 : 100,000 — about 0.00001 parts absorbed, 0.99999 parts non-absorbed.**

Ion Trapping of Aspirin

In the stomach, aspirin is non-ionized and absorbed into stomach cells, but once inside, the pH changes from 1.5 outside to 7.4 inside the cell.

So aspirin becomes ionized inside the cells and can't diffuse out of them again → gastric ulcer.



Clinical Significance of pKa

1. **Knowing the site of drug absorption from the GIT.**
2. **Treatment of drug toxicity:**
   - Toxicity with acidic drugs (e.g., aspirin): treat by alkalinization of urine — renders the drug more ionized in urine and less reabsorbable.
   - Toxicity with basic drugs (e.g., amphetamine): treat by acidification of urine renders the drug more ionized in urine and less reabsorbable.
3. **Ion trapping in breast milk:**
   - The pH of breast milk is 7, i.e., acidic relative to plasma (pH 7.4).
   - Basic drugs (pKa > 7.2) tend to be ionized, and thus trapped, inside breast milk more than acidic drugs.

Bioavailability

It is the fraction of unchanged drug reaching the systemic circulation following administration by any route.

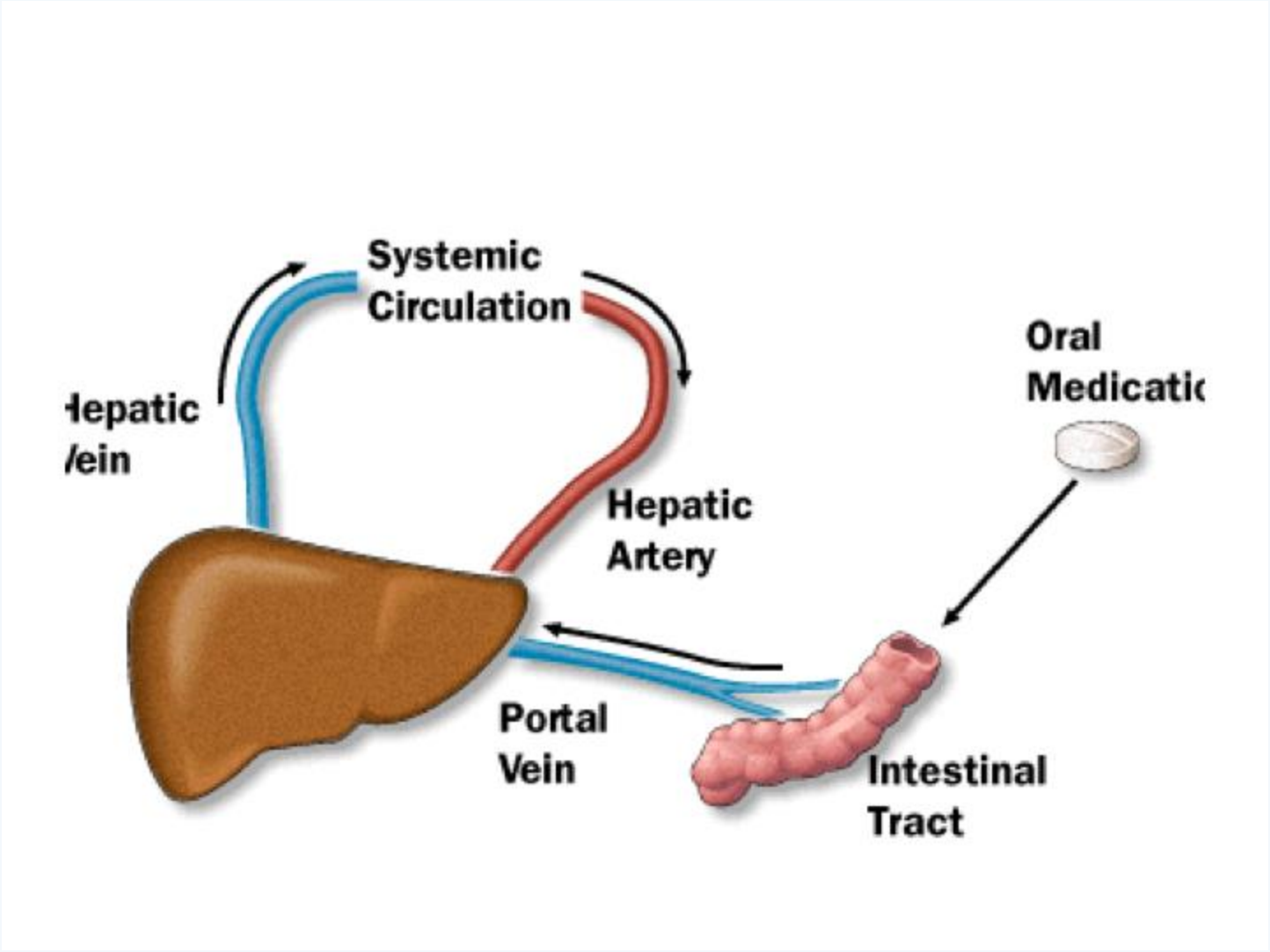
For the I.V. route, bioavailability is 100%, while for the oral route, it is less than 100% for two main reasons:

1. **Incomplete absorption**
2. **First-pass elimination**

First-Pass Hepatic Metabolism

- Biotransformation of a drug by liver or gut enzymes before it reaches systemic circulation, resulting in lower systemic effect (bioavailability) of the parent drug and diminished therapeutic effect.
- The first-pass effect may be **bypassed** if the drug is administered intravenously (IV), intramuscularly (IM), sublingually, as a suppository, by inhalation, or transdermally.
- Example: more than 90% of nitroglycerin is cleared during first-pass metabolism. Hence, it is primarily administered via the sublingual or transdermal route.

Hepatic Portal Circulation & First Pass



Distribution

Drug distribution is the reversible movement of a drug from the bloodstream to the extracellular fluid and/or to the cells of the tissues.

This movement depends on several factors.

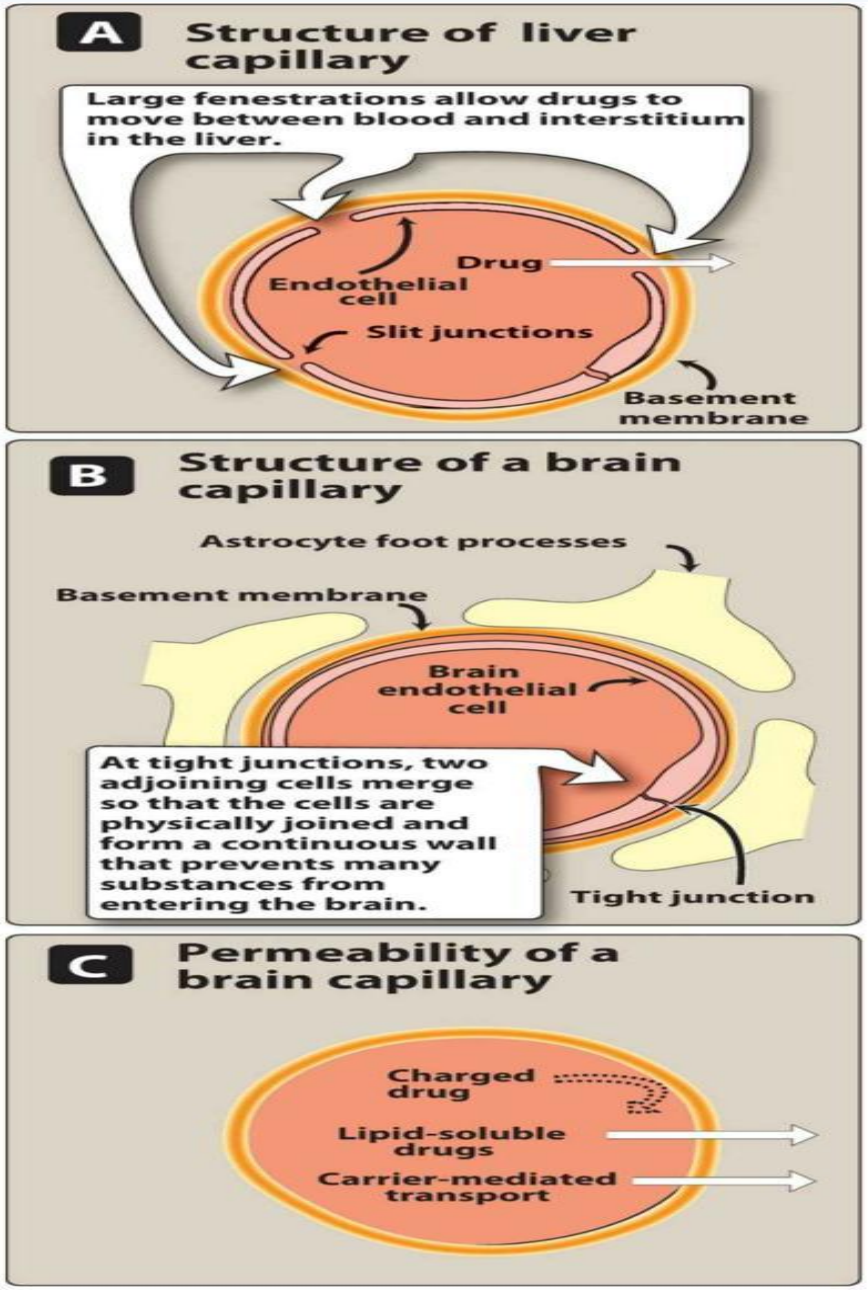
1 · Blood Flow

- Different organs have different blood flow rates— for example, the brain, liver & kidneys have ↑ blood supply compared with skeletal muscles, and adipose tissue has a very ↓ supply.
- Hypnosis produced by I.V. injection of Thiopental → rapid distribution to the CNS → anesthesia.
- Redistribution to peripheral tissues → regain consciousness.

2 · Capillary Permeability

- Depends on capillary structure.
- In the liver & spleen, a large part of the basement membrane is exposed (large discontinuous capillaries & slit junctions) — there is easy access for large plasma proteins.
- **In the brain, the reverse is true (continuous capillaries, no slit junctions) → Blood-Brain Barrier.**

The Blood-Brain Barrier



3 · Drug Structure

The chemical nature of a drug has a strong effect on its distribution.

Hydrophobic (lipophilic)

- Have no charge.
- Move through most biological membranes.

Hydrophilic (lipophobic)

- Have a −ve or +ve charge.
- Cannot pass through membranes, but can pass through slit junctions.

4 · Binding to Plasma & Tissue Proteins

Binding to plasma proteins:

At therapeutic concentrations in the plasma, many drugs exist mainly in bound form.

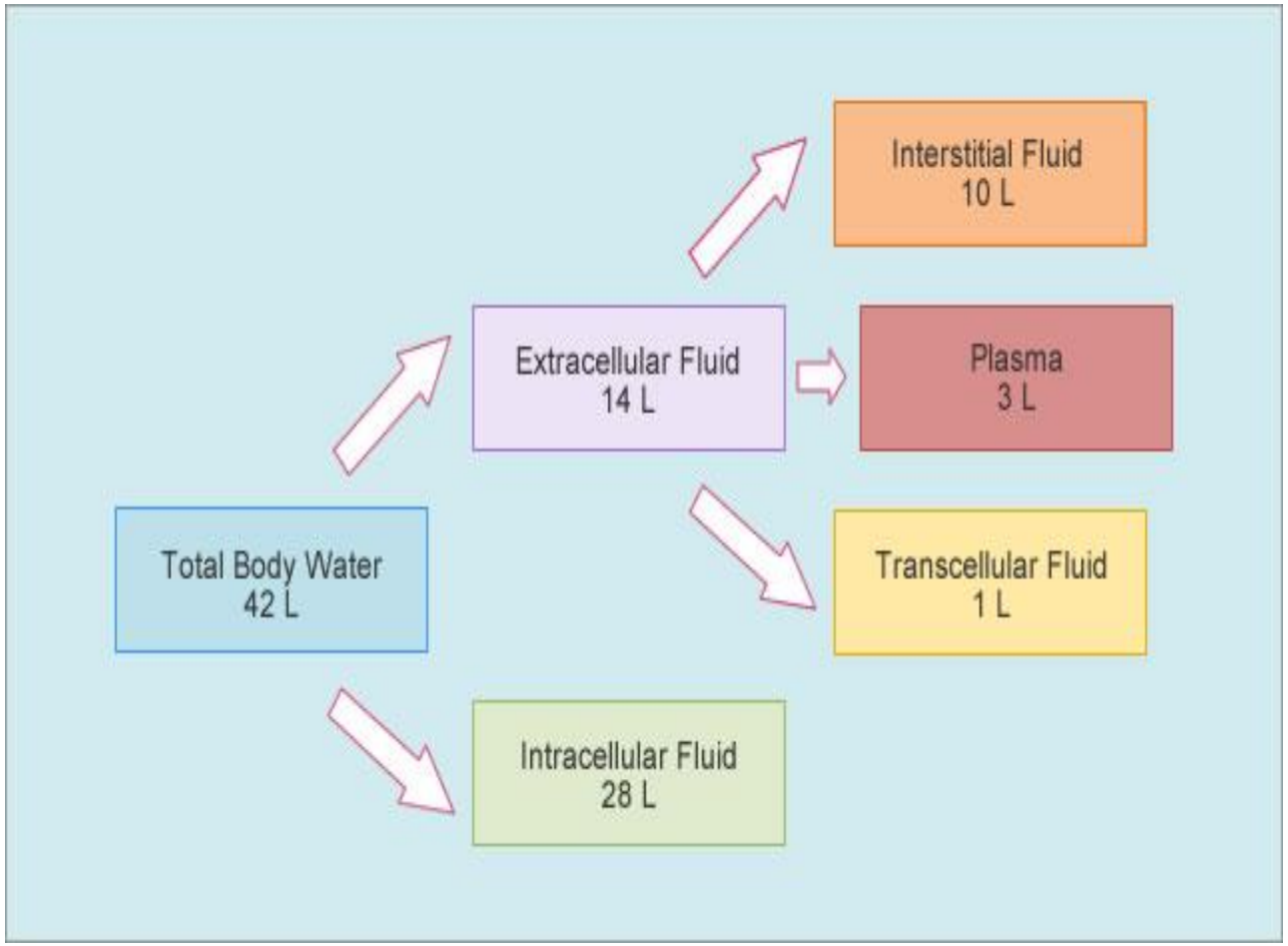
- The free fraction could be as low as 1%, with the remainder being bound to plasma proteins.
- It is the unbound form of the drug that is pharmacologically active.
- **↓ Vd (drug stays in plasma) — e.g., warfarin, phenytoin.**

Binding to Tissue Proteins

- Many drugs accumulate in tissues rather than in blood or E.C.F.
- Act as a reservoir for the drug.
- Prolong its action.
- Might be a source of toxicity.
- **↑ Vd (drug leaves plasma, accumulates in tissue) — e.g. Digoxin, chloroquine.**

Distribution of Drugs

Distribution into the water compartments in the body



1 · Plasma — 3 liters

If a drug has **a high molecular weight** or is **extensively protein-bound,** it is too large to pass through the slit junctions of the capillaries and, thus, is effectively trapped within the plasma (vascular) compartment.

Example: enoxaparin.

Water Compartments (continued)

2 · Extracellular water — 11 liters

A drug with low molecular weight but hydrophilic can pass through the endothelial slit junctions of the capillaries into the interstitial fluid. However, hydrophilic drugs cannot move across the lipid membranes of cells to enter the intracellular fluid.

Ex: Aminoglycoside, Amoxicillin.

3 · Intracellular water — 28 liters

A drug with low molecular weight and lipophilic character can move into the interstitium through the slit junctions and pass through the cell membranes into the intracellular fluid.

Ex: Ethanol, Diazepam, nitroglycerin.

Volume of Distribution

It is a hypothetical volume of fluid into which the drug is distributed at a concentration equal to that in plasma.

Calculation

$$Vd = \frac{\text{Total amount of the drug in the body}}{\text{Plasma conc of the drug (after distribution equilibrium)}} \ L$$

Clinical Significance

Determination of the site of drug distribution, e.g.:

- A total Vd < 5 L: the drug is confined to the vascular compartment and can be removed by dialysis.
- A total Vd 5–15 L: the drug is restricted to the ECF.
- A total Vd > 42 L: the drug is highly bound to tissue proteins and cannot be removed by dialysis.
- Calculation of the total amount of drug in the body by a single measurement of plasma concentration (from the equation).
- **Calculation of the loading dose (LD) needed to attain a desired plasma concentration (Cp): LD = Vd × Cp.**

Factors Affecting Vd

1. Lipid solubility

- Drugs with high lipid solubility or low rates of ionization have a high volume of distribution.
- Lipid-soluble drugs reach all compartments and may accumulate in fat.
- Lipid-insoluble drugs are mainly confined to plasma and interstitial fluids.

Factors Affecting Vd (continued)

2. Binding of drugs to plasma proteins

- Most drugs, when introduced into the body, are bound to plasma proteins.
- Albumin: the most important plasma protein — it can bind –ve or +ve charged drugs. Relatively non-selective binding.
  - –Weak acids bind to albumin, e.g., phenytoin & barbiturates.
  - –Weak bases bind to α-1-acid glycoprotein, e.g., propranolol & lignocaine.

Clinical Significance

- The pharmacological effect of the drug is produced through its free part, not its bound part (the bound part acts only as a reservoir from which the drug is slowly released).
- Binding of drugs to plasma proteins prolongs their effects.
- **When a drug has high plasma protein binding (e.g. 99% for warfarin), the free part that exerts the pharmacologic effect is 1%. Any small displacement of the bound part by another drug can lead to dramatic toxicity.**

Disease States & Drug Interactions

Many disease states (e.g. chronic liver disease, pregnancy, renal failure) can affect the level of albumin and the nature of plasma proteins, thus causing serious problems with some drugs.

Competition between drugs or with endogenous substances can lead, rarely, to clinically important drug interactions.

Example: sulfonamide can displace unconjugated bilirubin from binding to albumin, which might cause bilirubin encephalopathy in the newborn.

References

1. Lippincott's Illustrated Reviews Pharmacology, 6th & 7th edition

2. Mansoura Clinical Pharmacology, volume 1